MATHEMATICS ASSIGNMENT

MODULE 1

	Assignment questions
1.	Use Taylor's series method to find y at $x = 0.1,0.2,0.3$ considering terms upto the third degree given that $\frac{dy}{dx} = x^2 + y^2$ and $y(0) = 1$
2.	Use Picard's method to obtain the Second approximation to the solution of $\frac{dy}{dx} = x - y^2, x_0 = 0, y_0 = 1$ initially. hence compute $y(0.1)$correct to four decimal places
3.	Use fourth order Runge-Kutta method to find $y(0.1)$ for the equation $\frac{dy}{dx} = 3e^x + 2y, y(0) = 1$ taking $h = 0.1$
4.	Apply Milne's method to compute $y(1.4)$ correct to four decimal places given $\frac{dy}{dx} = x^2 + \frac{y}{2}$ and the following data: $y(1) = 2$, $y(1.1) = 2.2156$, $y(1.2) = 2.4649$, $y(1.3) = 2.7514$

MODULE 2

	Assignment questions
1.	Let A, B, C are fuzzy sets defined on universal set $X = \{a, b, c, d, e\}$ as follows $C = \left\{\frac{1}{a} + \frac{0.3}{b} + \frac{0.6}{c} + \frac{0.06}{d} + \frac{0.7}{e}\right\}$. calculate: i) (A∪B) ii) (B∪C) iii) (A∪C) iv) (B∩ C) v) check A∪(B∩ C) =(A∪B) ∩(A∪C)
2.	Let A be a fuzzy set defined on X where, $X = \{0,1,2,3,4,5\}$.Such that $A(x) = \frac{x}{x+2}$. Find α −Cut and Strong α-Cut for $\alpha = 0.2, 0.8, 1$.
3.	Let A, B, C are fuzzy sets as follows $A(x) = \frac{x}{x+5}, B(x) = \frac{x}{10}, C(x) = \frac{x}{x+10}$ and $x \in (0,1,2,3,4,5)$.Verify $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.
4.	Let A be the Fuzzy set defined on $X = \{-3, -2, -1, 0, 1, 2, 3, 4, 5\}$ and $A(x) = \frac{12-x}{15}$, $f(x) = x^2 + 2$ is a Crisp function in Which $x \epsilon X$ then by using Extension Principle.Find $f(A)$.

MODULE 3

	Assignment questions
1.	State and prove Cauchy's integral theorem.
2.	Evaluate $\int \frac{e^z}{z-i\pi}$ where C is the circle (a) $\|z\| = 2\pi$ (b) $\|z\| = \frac{\pi}{2}$.
3.	Find the residues of the function $f(z) = \frac{z}{(z+1)(z-2)^2}$ at $(i)\ z = -1\ (ii)\ z = 2$.
4.	Find the bilinear transformation which map the points $z = 1, i, -1$ into $w = 2, i, -2$. Also find the invariant points of the transformation.